

to provide either mechanical structure or electrical connection. Some of these materials like silicon and gallium arsenide can also be used as active as well as passive materials.

Active materials. These materials are essential to the sensing process in microelectronics, photosensitive, piezoelectric, magnetoresistive and chemoresistive films. Microsensor materials in the form of thin or thick films play an active role in the sensing system. These devices are fabricated using chemical vapour deposition (CVD) or low-pressure chemical vapour deposition (LPCVD) and special techniques like electrochemical deposition.

Silicon. Elemental silicon is not found in nature, but occurs in compounds like oxides and silicates. Silicon is abundant, relatively inexpensive and exhibits a number of physical properties that are useful for sensor application. It is possible to deposit layers of materials with the desired properties on a silicon substrate. Single-crystalline silicon is the most widely used semiconductor material.

Polysilicon. Polycrystalline layers may be formed by vacuum deposition onto an oxidised silicon wafer with an oxide. Polysilicon structures may be doped with boron or other elements by ion implantation or other techniques to reach the required conductivity. The temperature coefficient of the resistance may be changed over a wide range, positive or negative—through selective doping. Polysilicon resistors have a long-term stability.

Other semiconductors. There is a wide range of compound semiconductors available to prepare heterostructures with unique properties. Gallium-arsenide (GaAs) and indium-antimonide (InSb) are widely used in electronic components.

GaAs. Gallium-arsenide is used in devices such as infrared light-emitting diodes, laser diodes, microwave monolithic integrated circuits (ICs) and solar cells. It is also used in optical fibre temperature sensors. A study shows that some electronic properties of gallium-arsenide are superior to those of silicon. Gallium-arsenide transistors function at frequencies above 250GHz. Due to the superior properties of GaAs, these are widely used in mobile

Some key sensor players

- Analog Devices Inc.
- Custom Sensors and Technologies Inc.
- Delphi Automotive PLC
- Infineon Technologies AG
- Omron Corp.
- Qualcomm Technologies
- Siemens Healthcare Diagnostics
- STMicroelectronics
- Vishay InterTechnology Inc.
- Wilcoxon Research Inc.

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phones, satellite communications and radar systems. The highly sensitive GaAs piezoelectric sensors are also used for biological detection.

InSb. It is useful for magnetic sensing devices such as Hall Effect sensors and magnetic resistors. InSb magnetoresistors are used as position sensors in automotive applications. InSb materials are also used for infrared imaging.

Plastics. Plastics are widely used in electronic and electrical components and assemblies. Since plastics are insulators, these are used in a variety of applications where insulation properties are needed. Polymers are also used as radiation detectors and chemical sensors.

Metals. Physical properties and mechanical processing of metals are taken into account while designing sensors. Copper has excellent thermal and electrical properties, but it is difficult to machine. Aluminium is used as an alternative in some cases. Metals are used in magnetic sensors. Precious metals like gold, silver, platinum, rhodium and palladium are widely used in sensor devices for automobiles, RFID tags, mobile phones and PCs.

Ceramics. Ceramics are widely used in sensor fabrication. These possess common properties including structural strength, light weight, thermal stability, electrical insulation and ability to bond with other materials. They do not react with oxygen and thus do not create oxides. Many manufacturers use ceramics as sensor substrates.

Sensor fabrication

Microsensor technology uses the basic fabrication steps followed in conven-

tional silicon planar IC technique and some additional steps. At present, complementary metal-oxide semiconductor (CMOS) is the most common technology used in microsensors. Microsensors are designed and fabricated using commercial CMOS IC processes with subsequent bulk-micromachining technology. The exact steps are different from sensors to sensors.

Encapsulation. The chip must be protected from the atmosphere. Photoresist or silicon nitride materials are often used to cover the sensing area. LPCVD or CVD process is used to deposit silicon nitride layer, which acts as a barrier against water. The next step is IC encapsulation. That includes sealing the IC in a plastic resin or metal case. This process protects the silicon device from the surrounding environment, and may not be always required in some MEMS devices where the atmosphere is used to transmit the measured quantity.

Deposition. Some sensors, especially MEMS devices, are required to deposit thin and thick film materials providing the sensing surface with the required properties. For example, sensitivity to thermal radiation is given by coating with nichrome. The film can be locally etched using lithography and wet chemical etching processes. Dry physical etching and laser processing can also be used.

The scope and future outlook

With micro and nano technologies, sensors can be made to fit almost anywhere in consumer devices, robots, automobiles and even human bodies. Use of intelligent sensors is also increasing in counter-terrorism, cargo tracking, biometrics among other applications. Latest sensors are used in automobiles to prevent impending crash and determine the type of airbags to be fired, and force and speed of their deployment. The use of MEMS in medical applications, including implantable devices and handheld devices for diagnostics and monitoring systems, is on the rise. Looking forward, with advancements in technology, the new wave of sensors including IoTs and wearables is going to revolutionise the electronics industry in years to come. **EFY**